



고려대학교
KOREA UNIVERSITY



AI Security

- Lecture 07. Convolutional Neural Networks

Artificial Intelligence Research (AIR) LAB

<https://air.korea.ac.kr/>

School of Cybersecurity

Korea University

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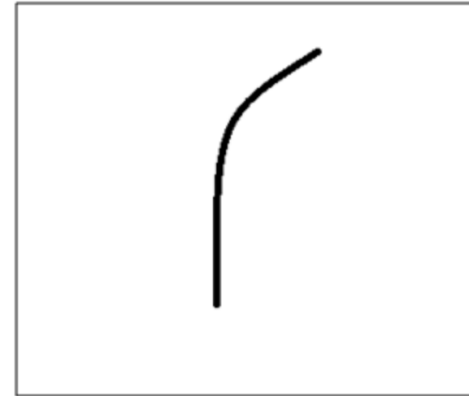
2025. Spring Semester

Convolution

필터 Filter
(커널 Kernel)

0	0	0	0	0	30	0
0	0	0	0	30	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	0	0	0	0

Pixel representation of filter



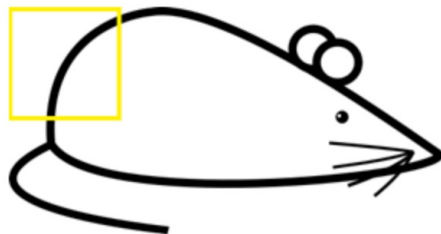
Visualization of a curve detector filter

학습대상
이미지



Original image

Convolution



Visualization of the filter on the image



Visualization of the receptive field

0	0	0	0	0	0	30
0	0	0	0	50	50	50
0	0	0	20	50	0	0
0	0	0	50	50	0	0
0	0	0	50	50	0	0
0	0	0	50	50	0	0
0	0	0	50	50	0	0

Pixel representation of the receptive field

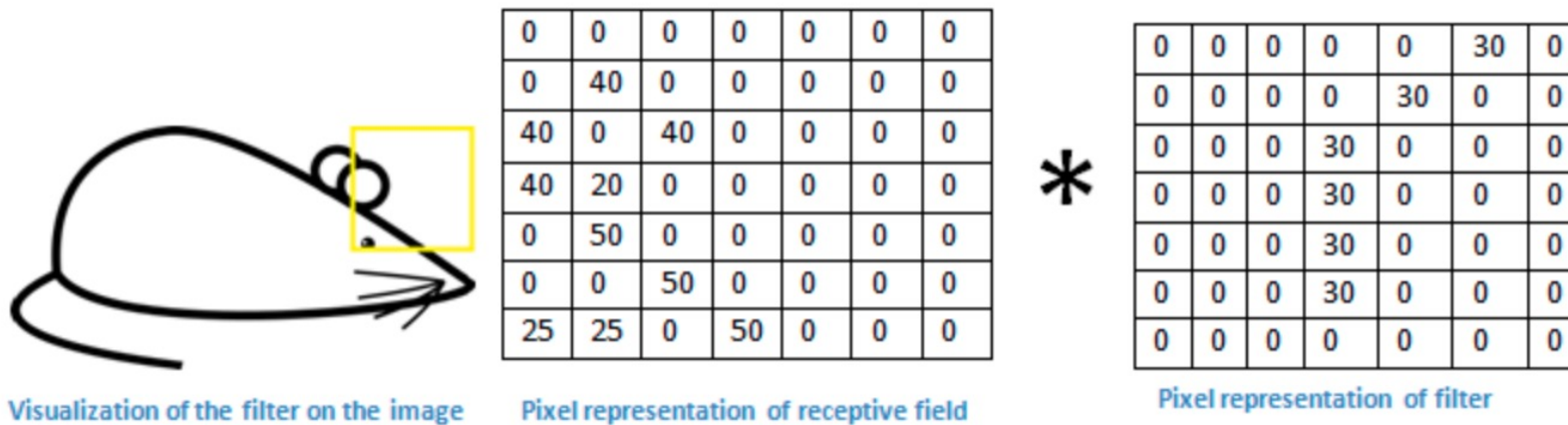
*

0	0	0	0	0	30	0
0	0	0	0	30	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	0	0	0	0

Pixel representation of filter

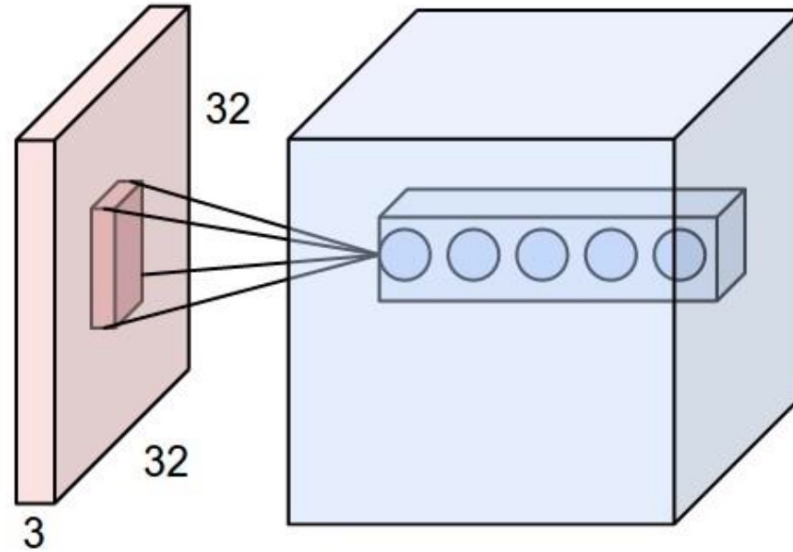
Multiplication and Summation = $(50*30)+(50*30)+(50*30)+(20*30)+(50*30) = 6600$ (A large number!)

Convolution



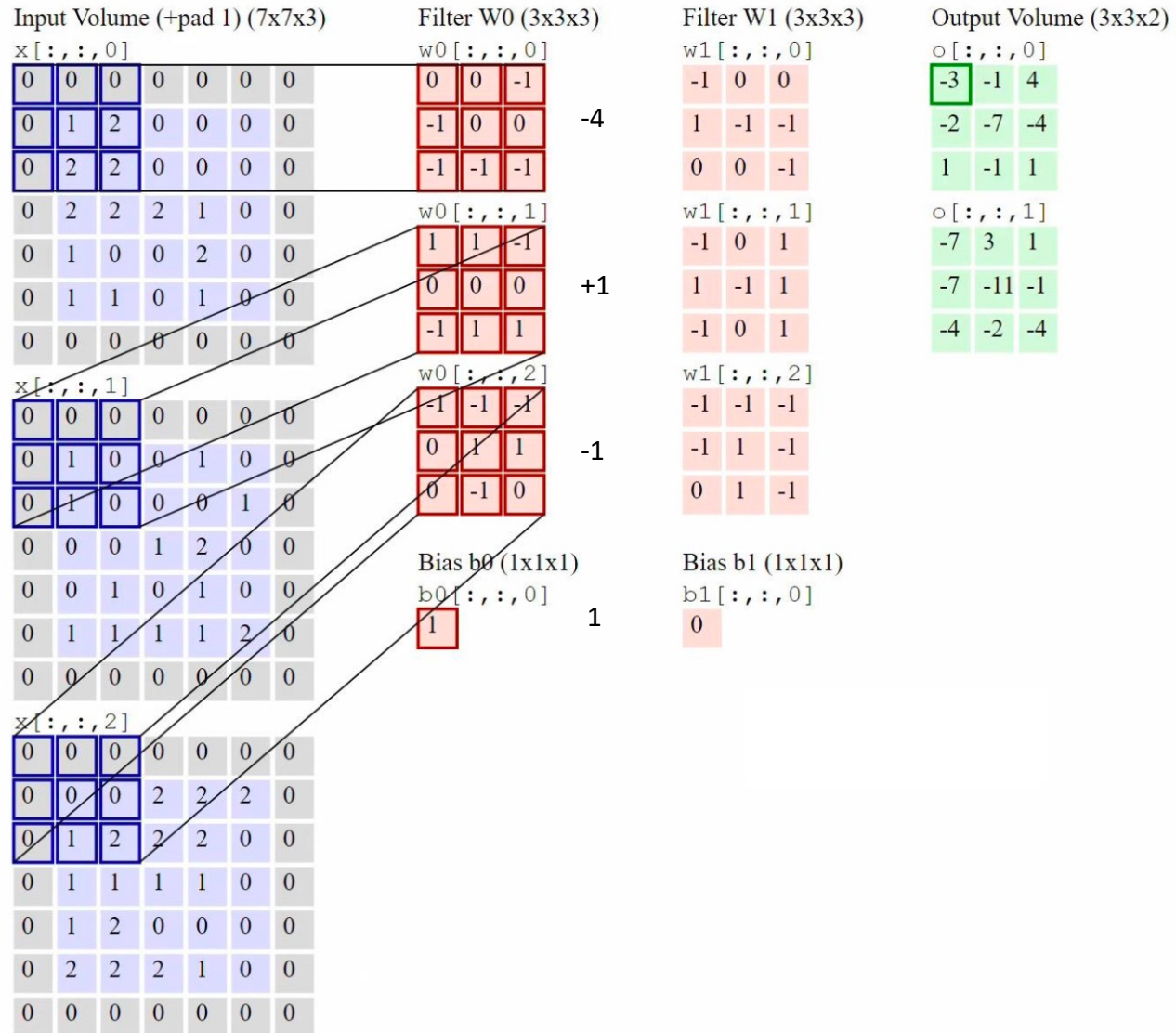
Multiplication and Summation = 0

Terminology

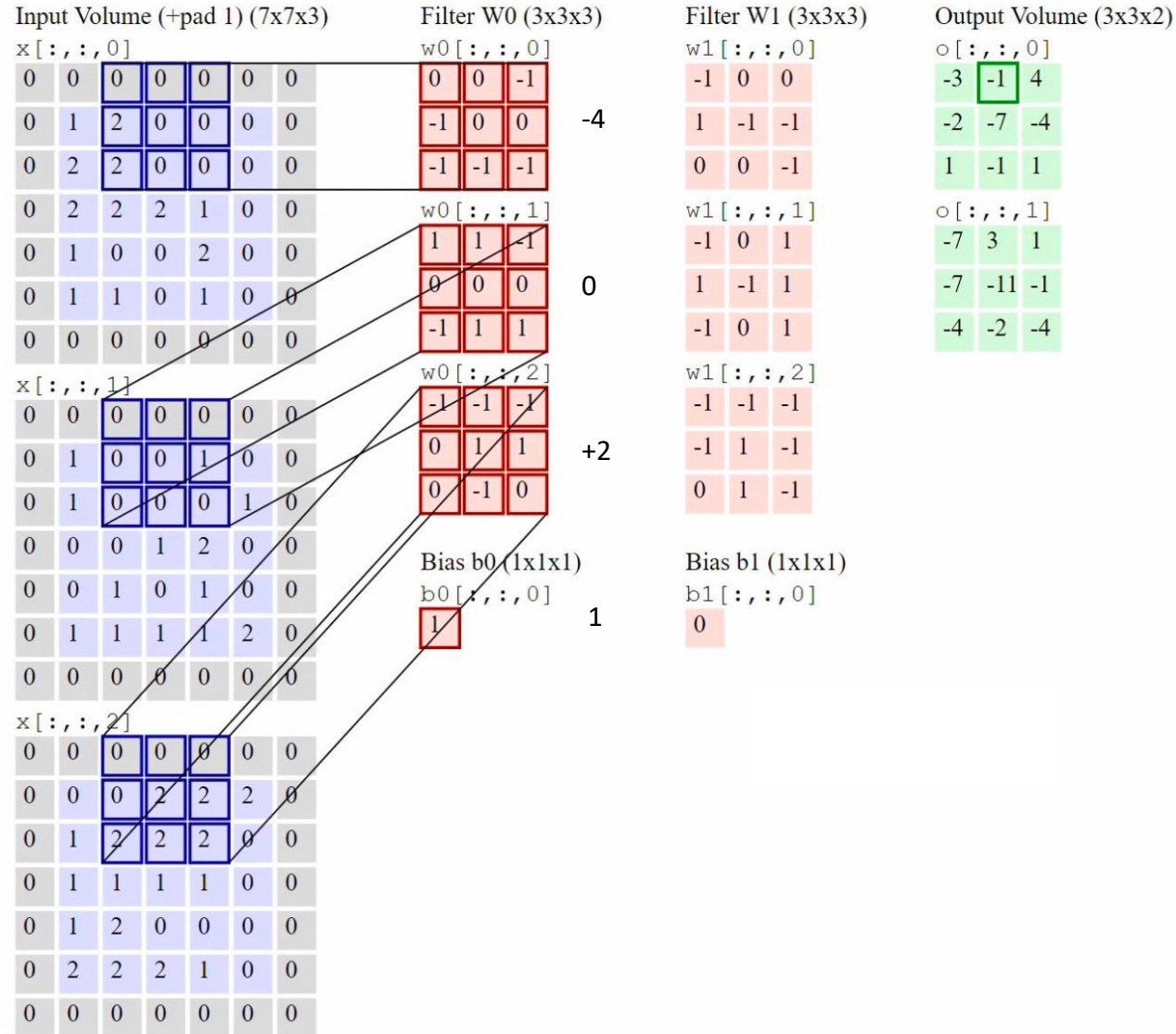


- **Depth:** number of filters
- **Stride:** filter step size (when we “slide” it)
- **Padding:** zero-pad the input

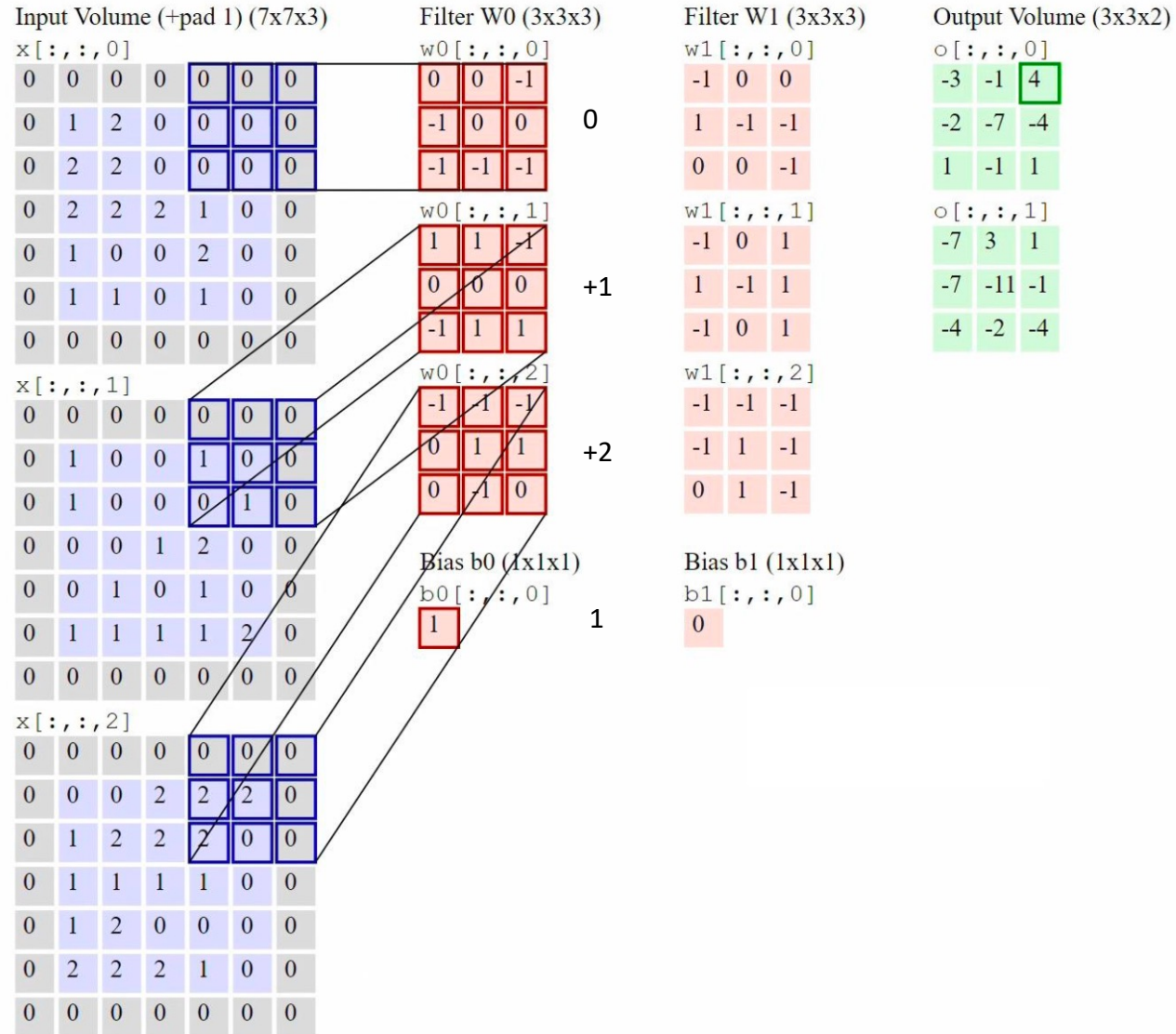
Convolution (Padding 1, Stride 2)



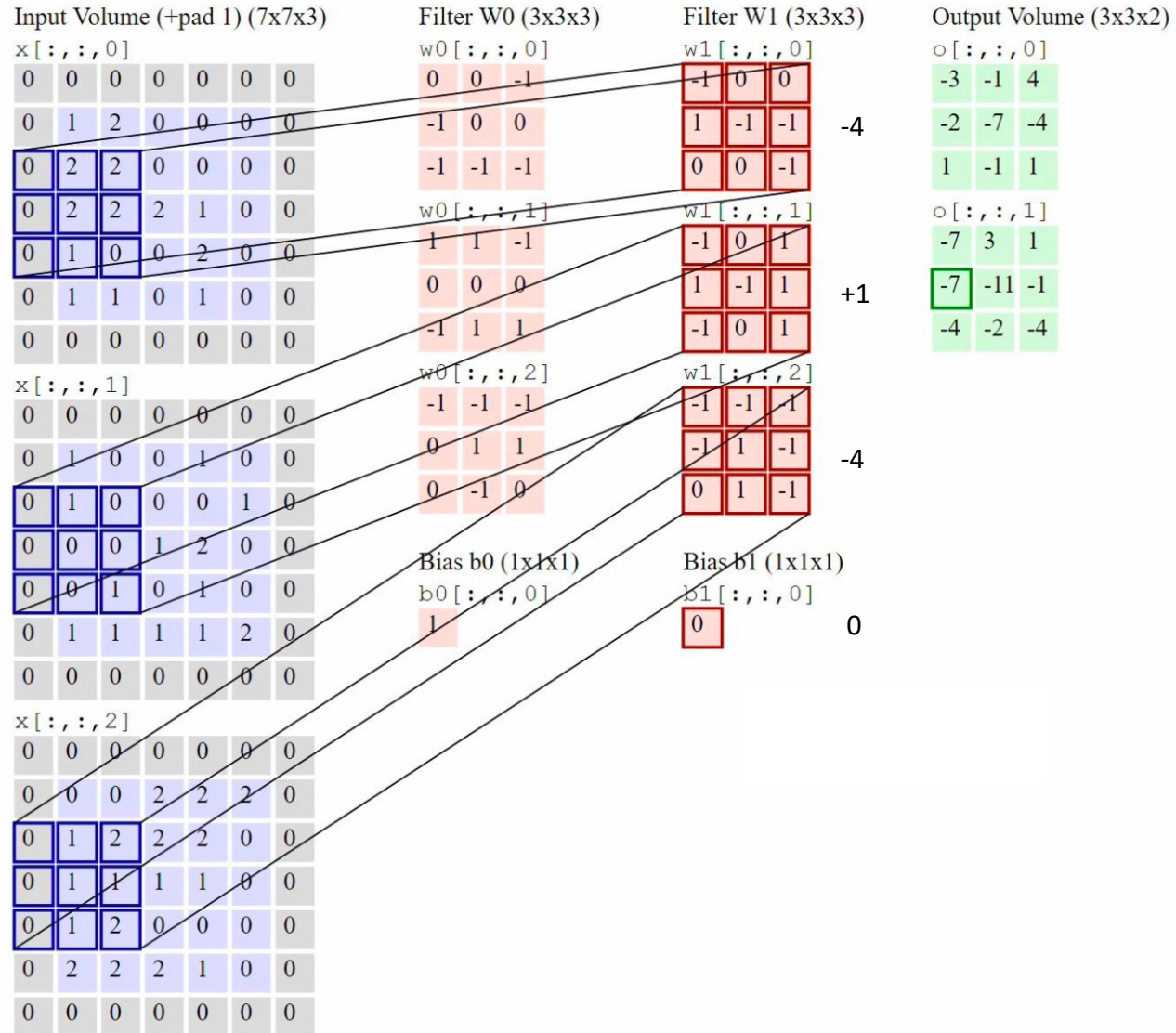
Convolution (Padding 1, Stride 2)



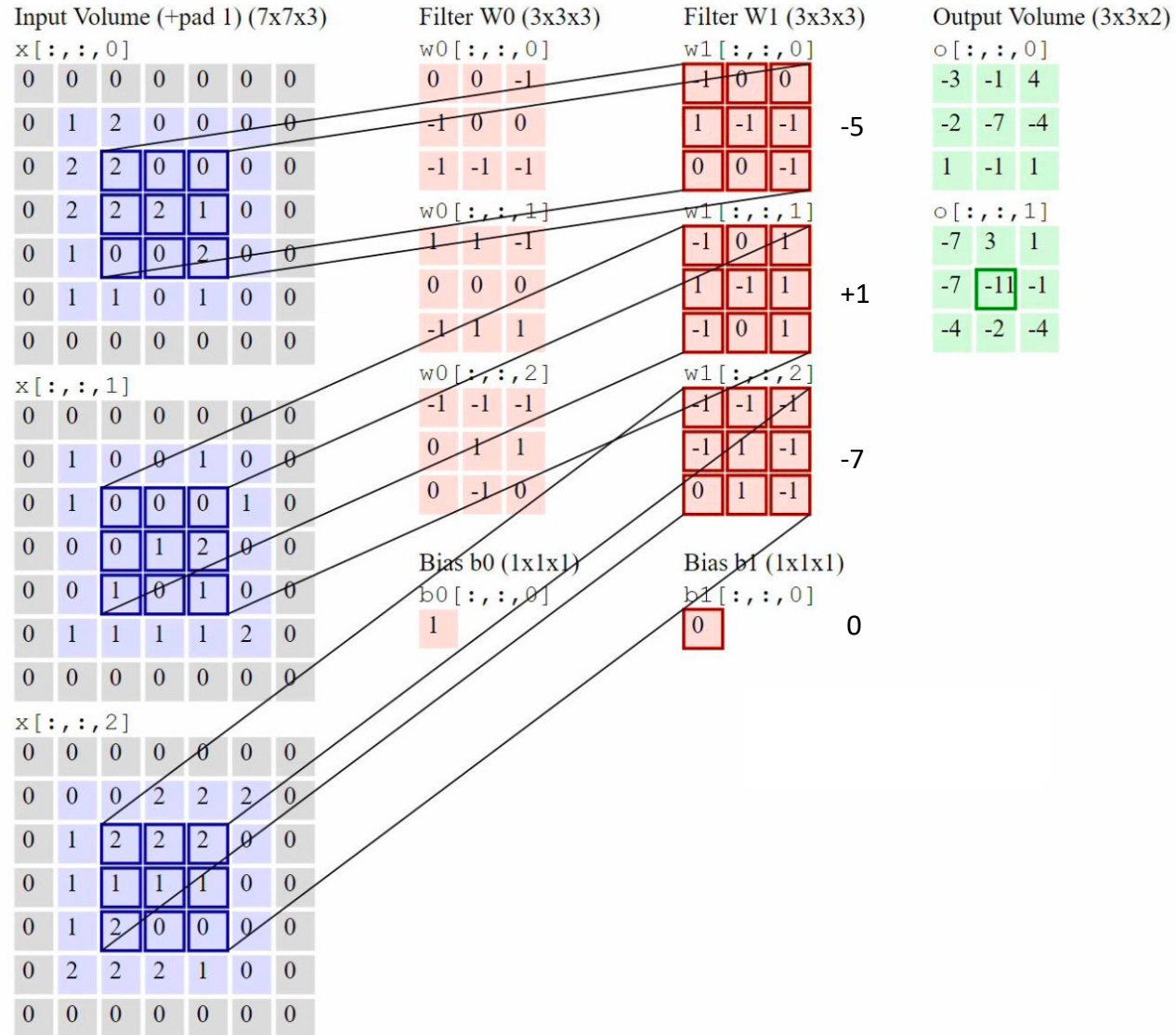
Convolution (Padding 1, Stride 2)



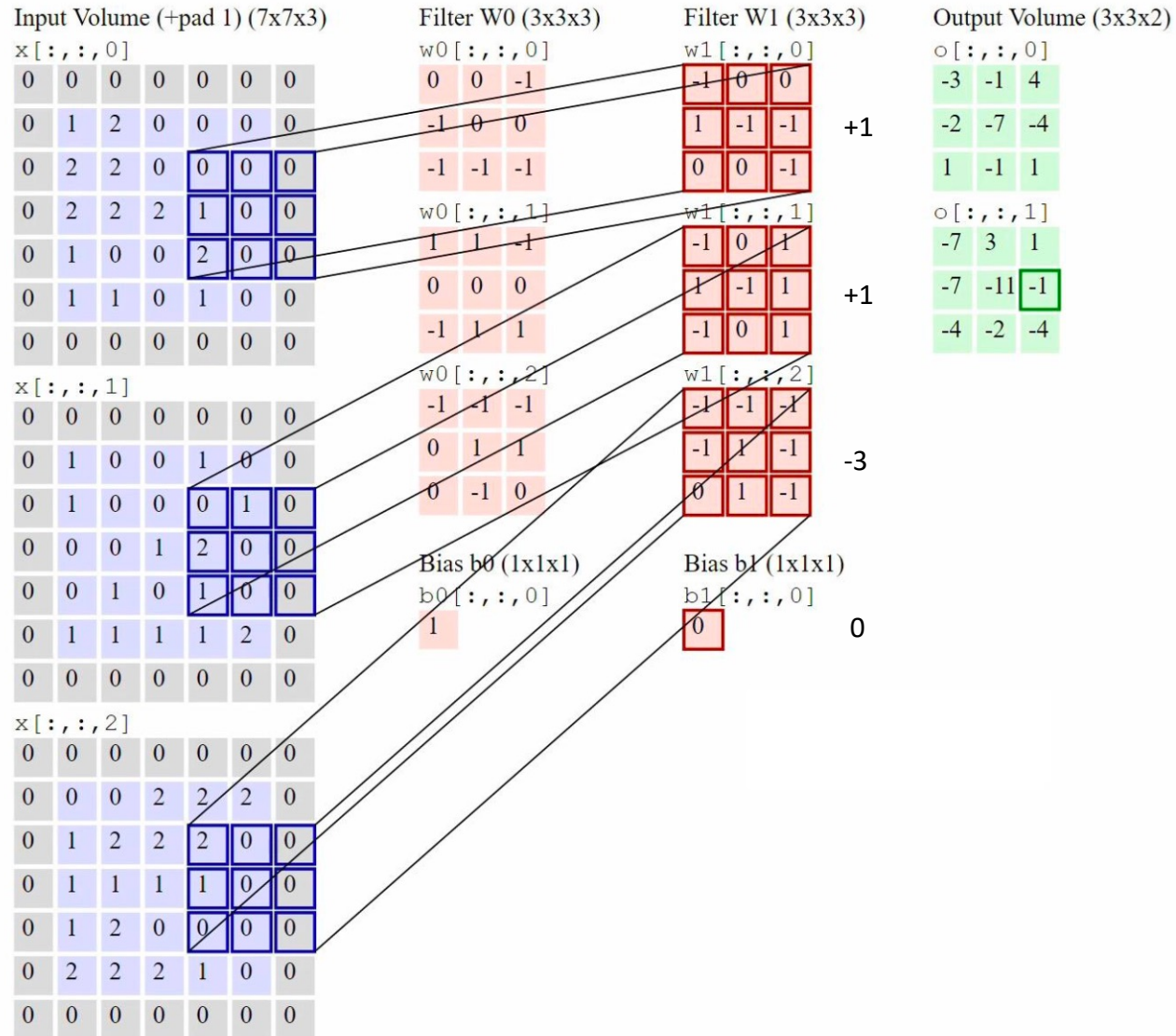
Convolution (Padding 1, Stride 2)



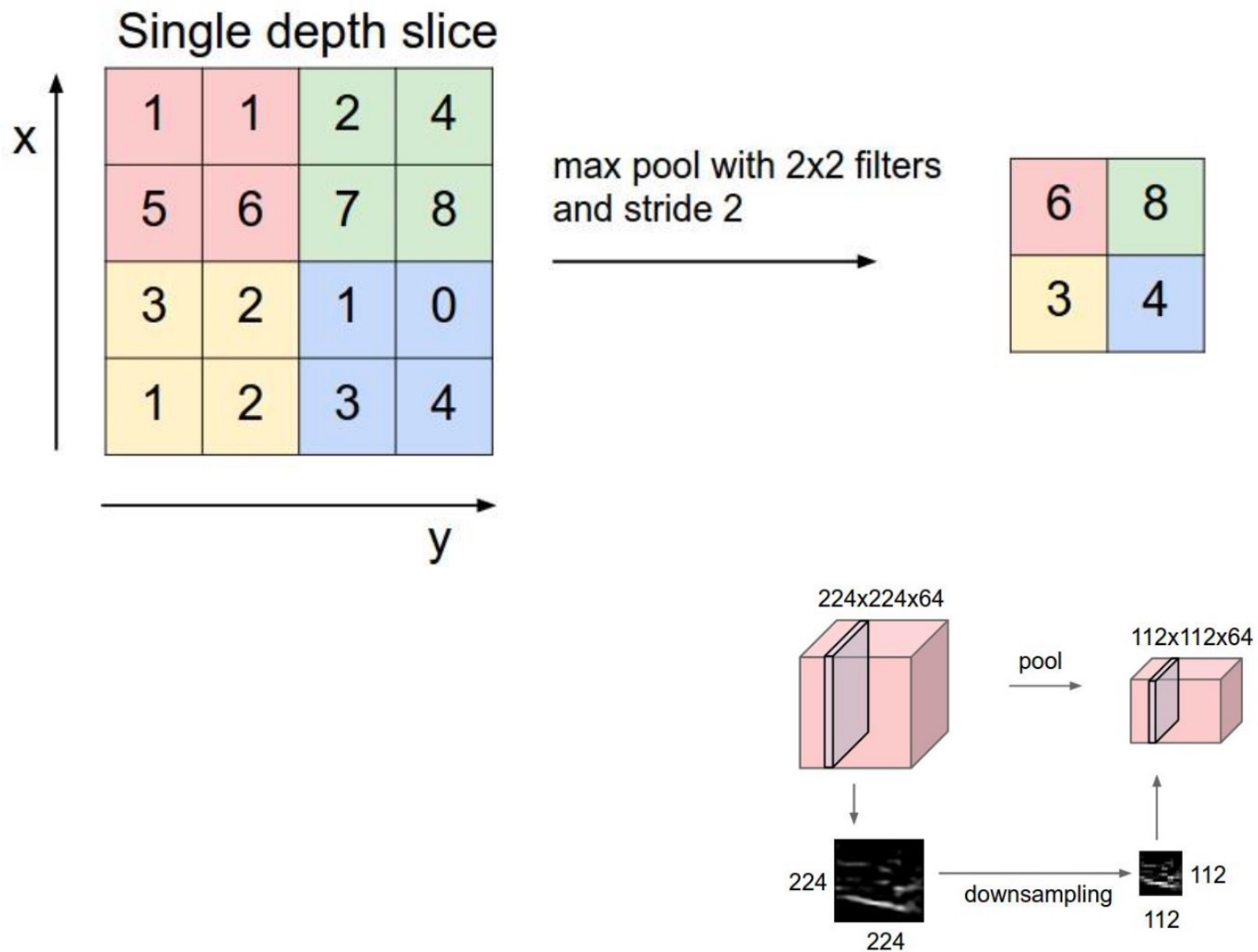
Convolution (Padding 1, Stride 2)



Convolution (Padding 1, Stride 2)

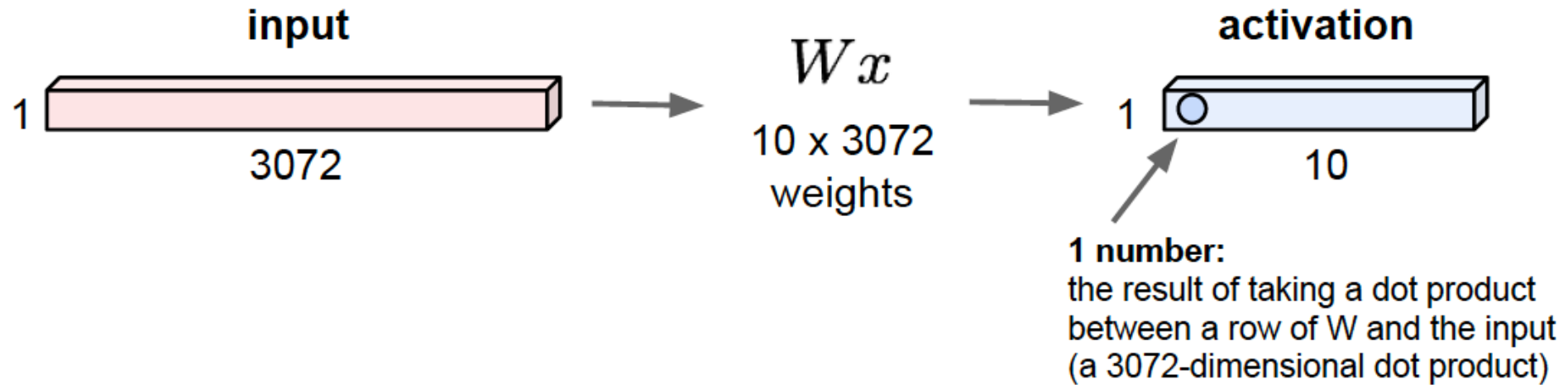


Pooling

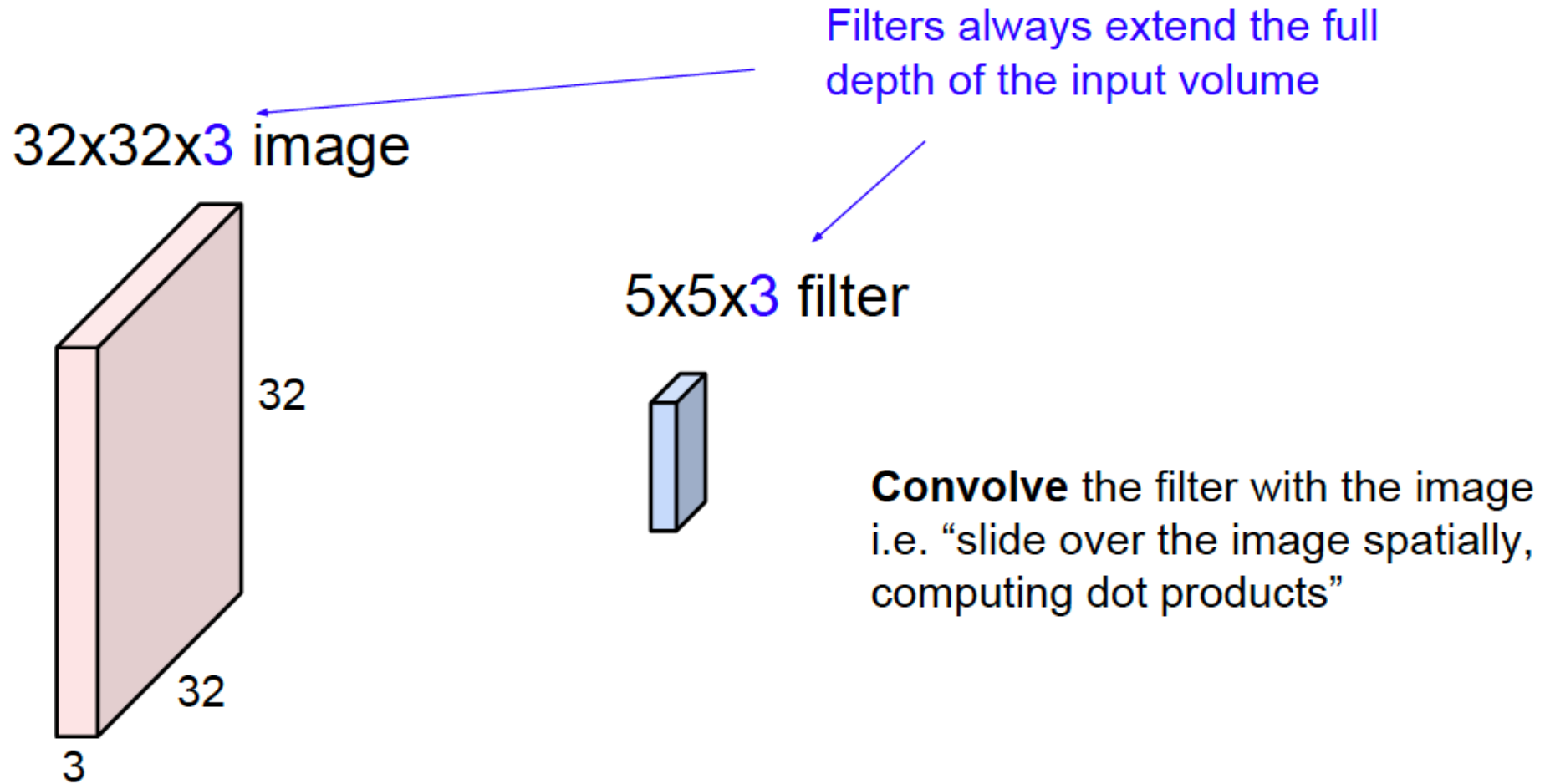


FC (Fully Connected) Layer

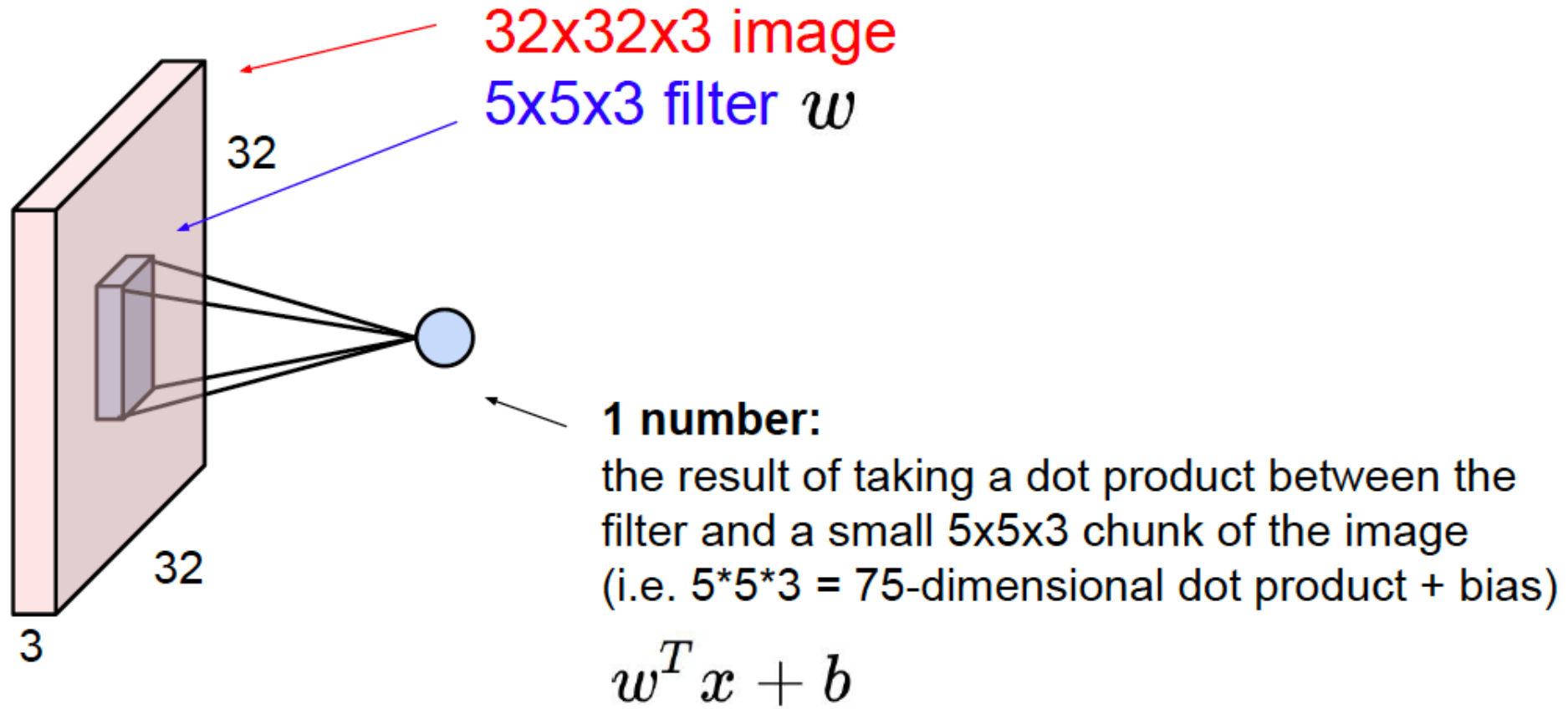
32x32x3 image -> stretch to 3072 x 1



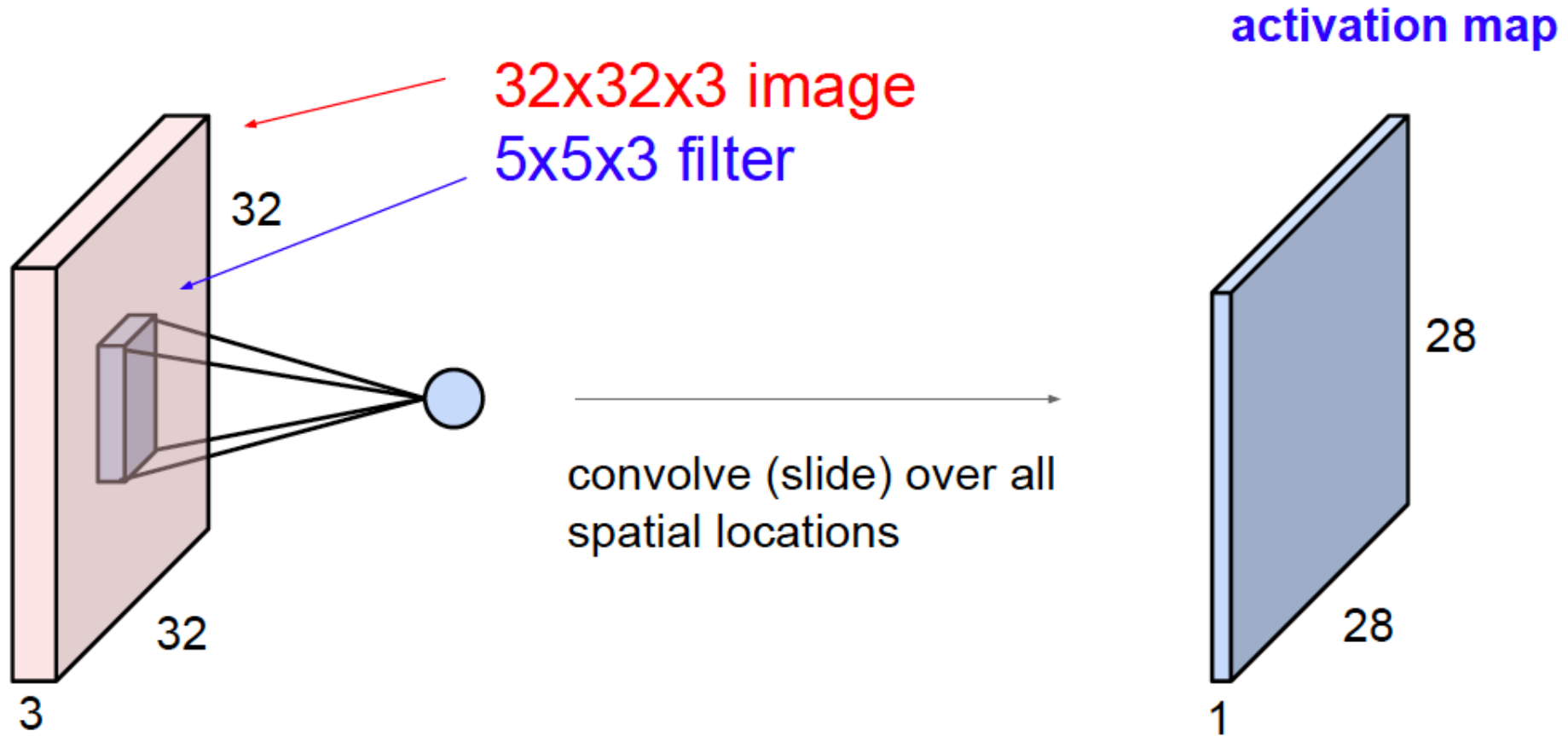
Convolution Layer



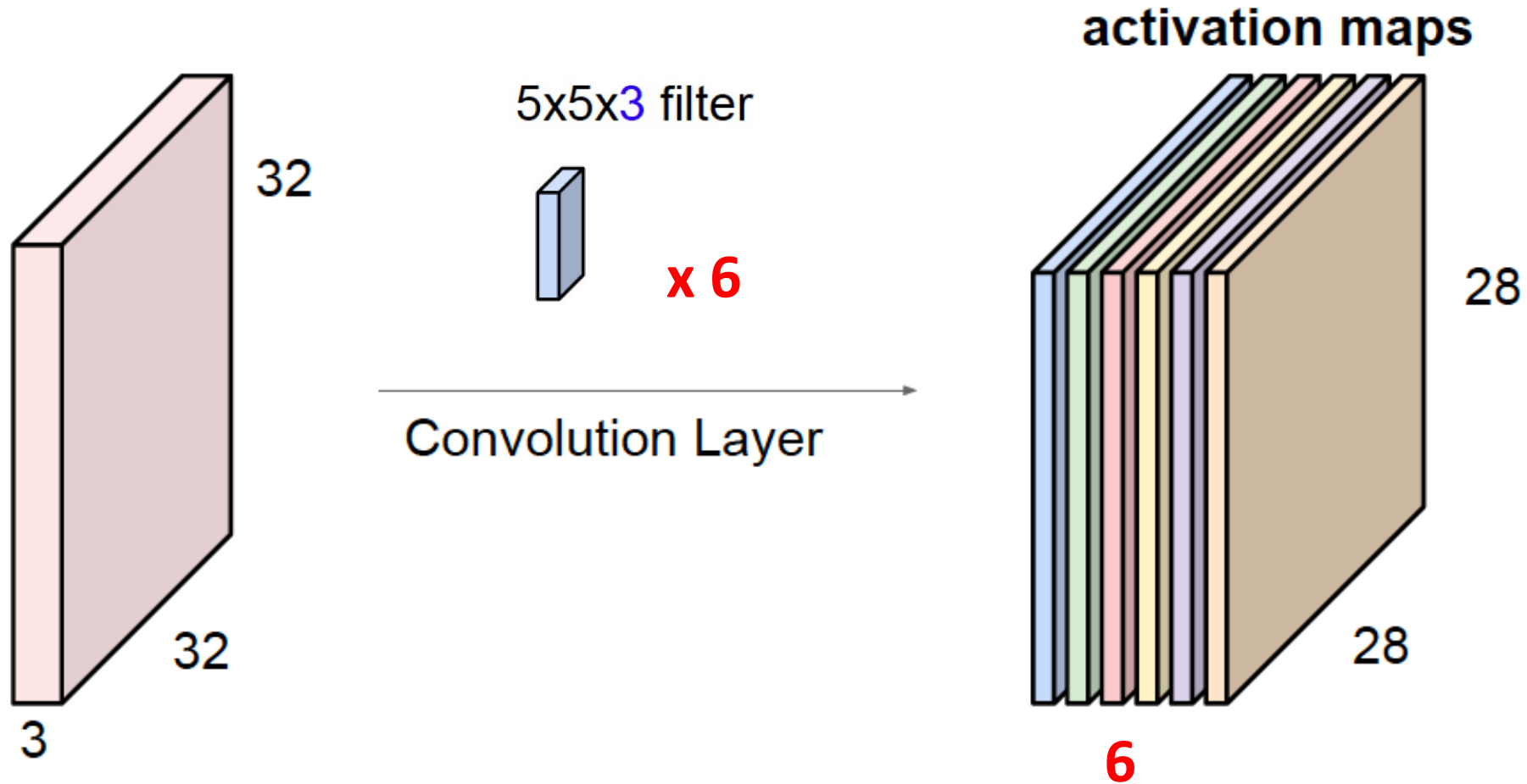
Convolution Layer



Convolution Layer

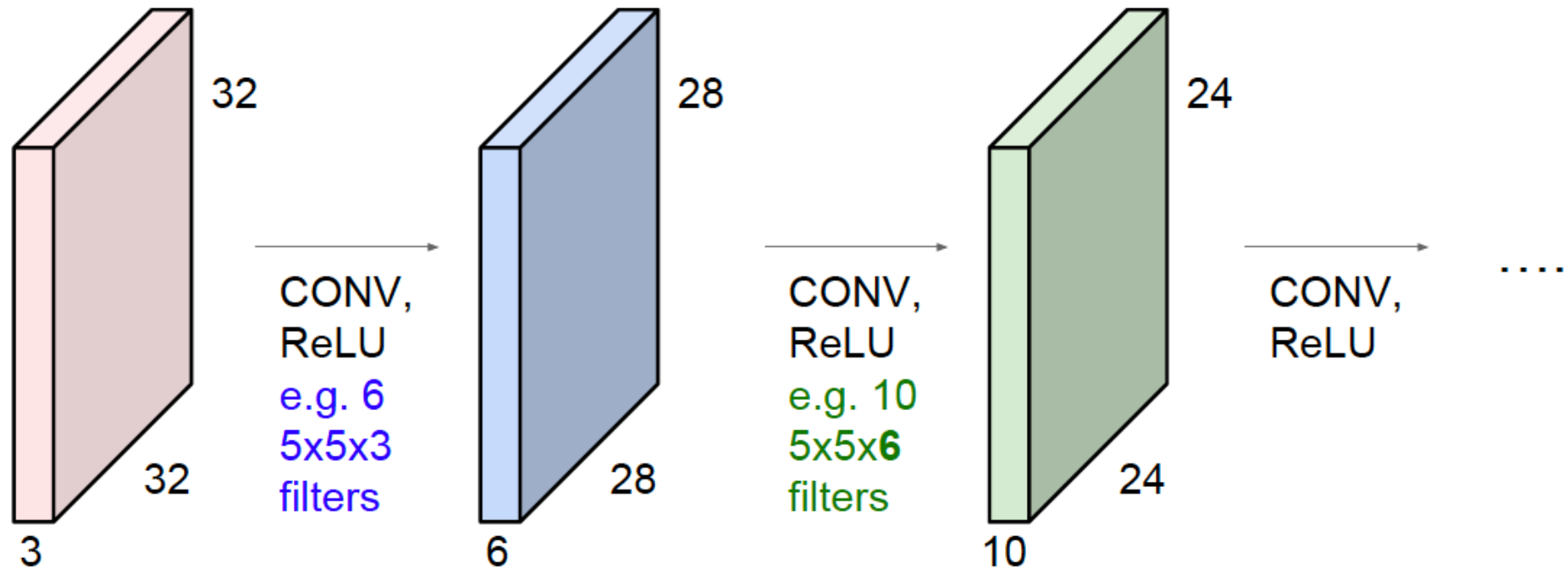


Convolution Layer

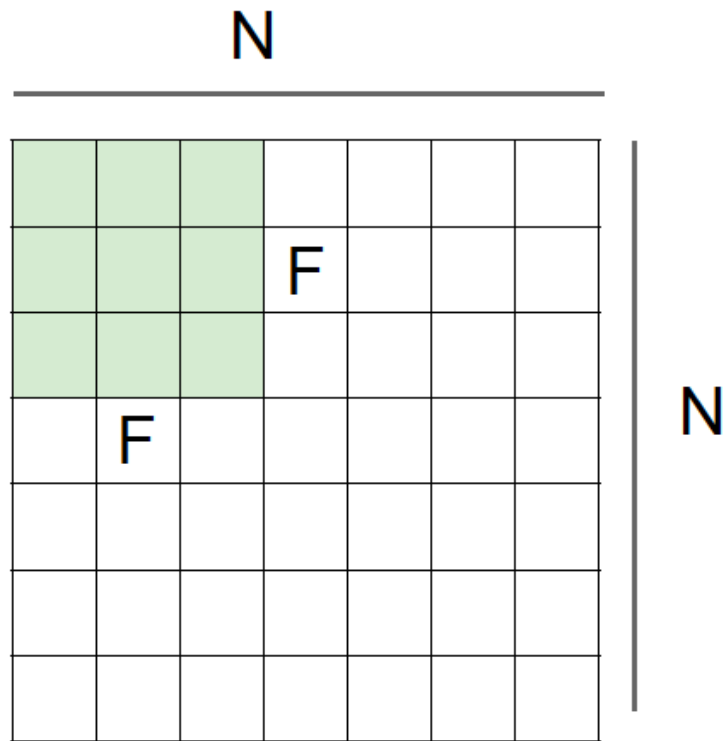


We stack these up to get a “new image” of size 28x28x6!

CNN



Output Size



Output size:
 $(N - F) / \text{stride} + 1$

e.g. $N = 7, F = 3$:

stride 1 $\Rightarrow (7 - 3) / 1 + 1 = 5$

stride 2 $\Rightarrow (7 - 3) / 2 + 1 = 3$

stride 3 $\Rightarrow (7 - 3) / 3 + 1 = 2.33 \therefore \backslash$

Output Size with Zero-Padding

0	0	0	0	0	0			
0								
0								
0								
0								

e.g. input 7x7

3x3 filter, applied with **stride 1**

pad with 1 pixel border => what is the output?

7x7 output!

in general, common to see CONV layers with stride 1, filters of size $F \times F$, and zero-padding with $(F-1)/2$. (will preserve size spatially)

e.g. $F = 3 \Rightarrow$ zero pad with 1

$F = 5 \Rightarrow$ zero pad with 2

$F = 7 \Rightarrow$ zero pad with 3

Output Size

Input volume: **32x32x3**

10 5x5 filters with stride 1, pad 2

Output volume size: ?

Output Size

Input volume: **32x32x3**

10 **5x5** filters with stride **1**, pad **2**

Output volume size:

$(32 + 2 * 2 - 5) / 1 + 1 = 32$ spatially, so

32x32x10

No. of Parameters

Input volume: **32x32x3**

10 5x5 filters with stride 1, pad 2

Number of parameters in this layer?

No. of Parameters

Input volume: $32 \times 32 \times 3$

10 5×5 filters with stride 1, pad 2

Number of parameters in this layer?

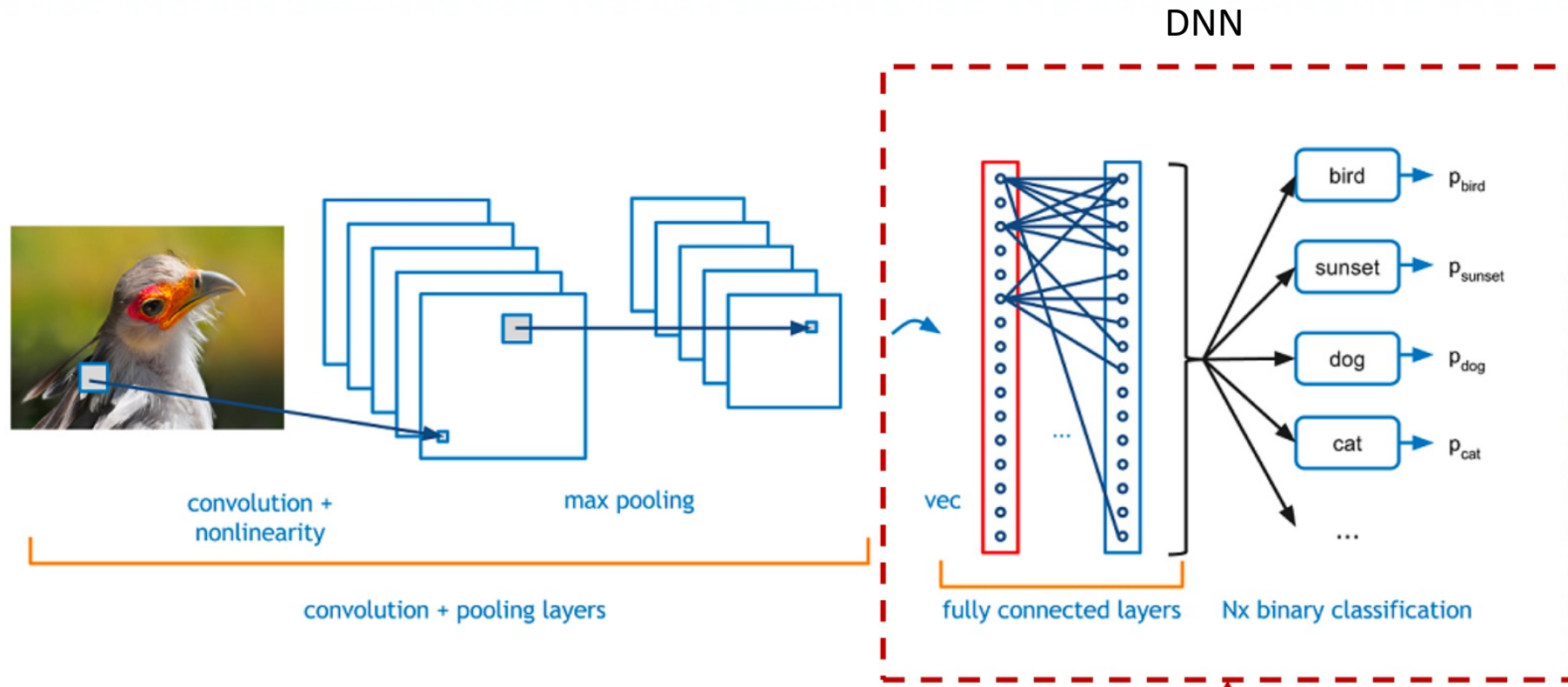
each filter has $5 \times 5 \times 3 + 1 = 76$ params

$\Rightarrow 76 \times 10 = 760$

Spatial Dim & No. Parameters

- Input volume: $W1 \times H1 \times D1$
- Filter (kernel)
 - No of filters K
 - Spatial extent F
 - Stride S
 - Amount of zero padding P
- Output volume: $W2 \times H2 \times D2$
 - $W2 = (W1 - F + 2P) / S + 1$
 - $H2 = (H1 - F + 2P) / S + 1$
 - $D2 = K$
- With weight sharing,
 - $(F \times F \times D1) \times K$ weights
 - K biases

CNN Architecture



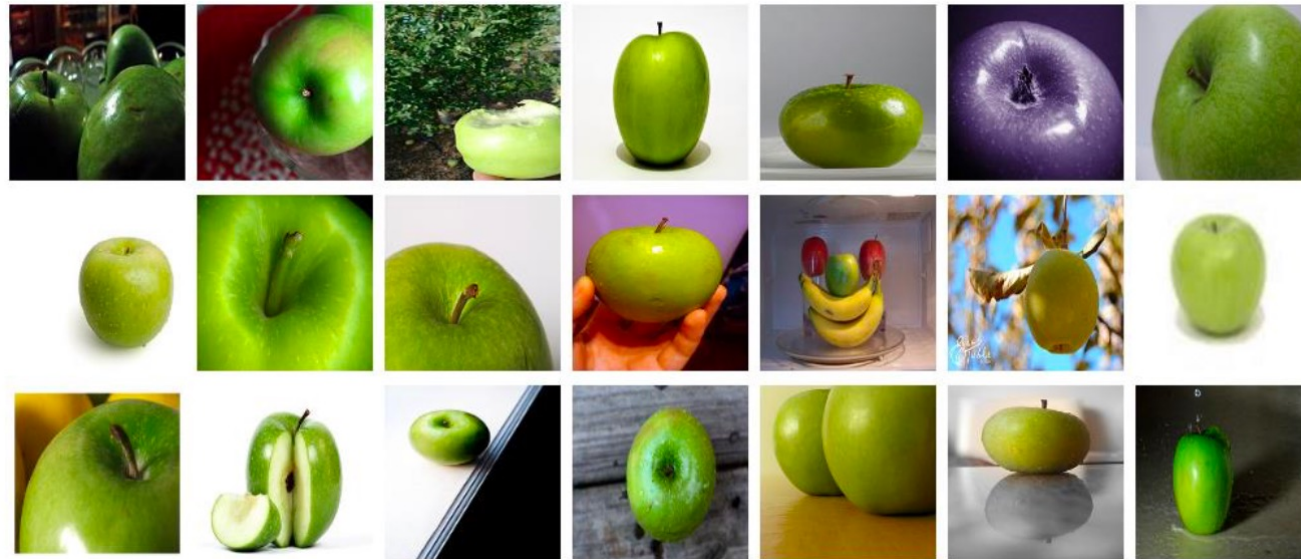
응용에 따라 Convolution 부분이나 DNN 구조를 바꿀 수 있음

ImageNet Data

Dataset of 14+ million images of 21,841 categories

Category “Fruit”: 180,000 images

- 1206 Granny Smith apples



ILSVRC : ImageNet Large Scale Visual Recognition Challenge

ILSVRC Challenge

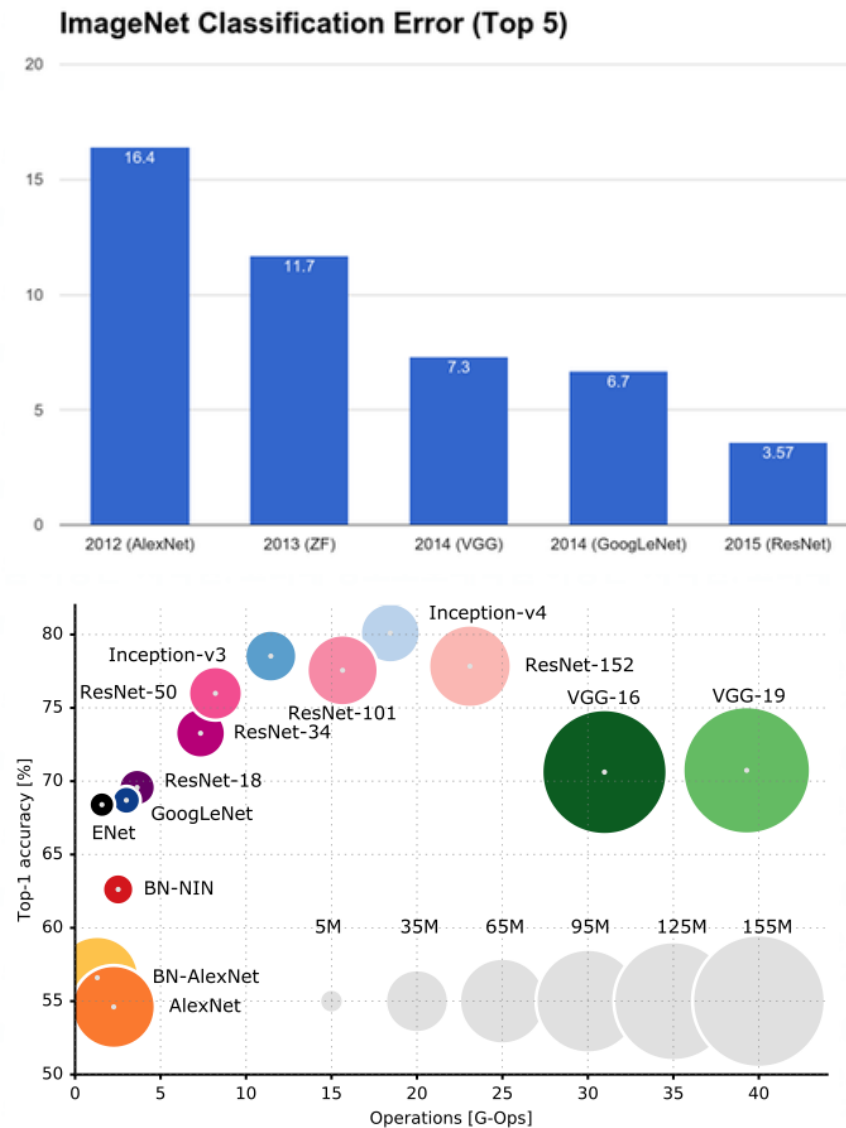
Top 5 error rate:

- Can make 5 guesses to get the correct label

Image classification			
Steel drum 	<u>Steel drum</u> Folding chair Loudspeaker	Scale T-shirt <u>Steel drum</u> Drumstick Mud turtle	Scale T-shirt Giant panda Drumstick Mud turtle
Ground truth	Accuracy: 1	Accuracy: 1	Accuracy: 0

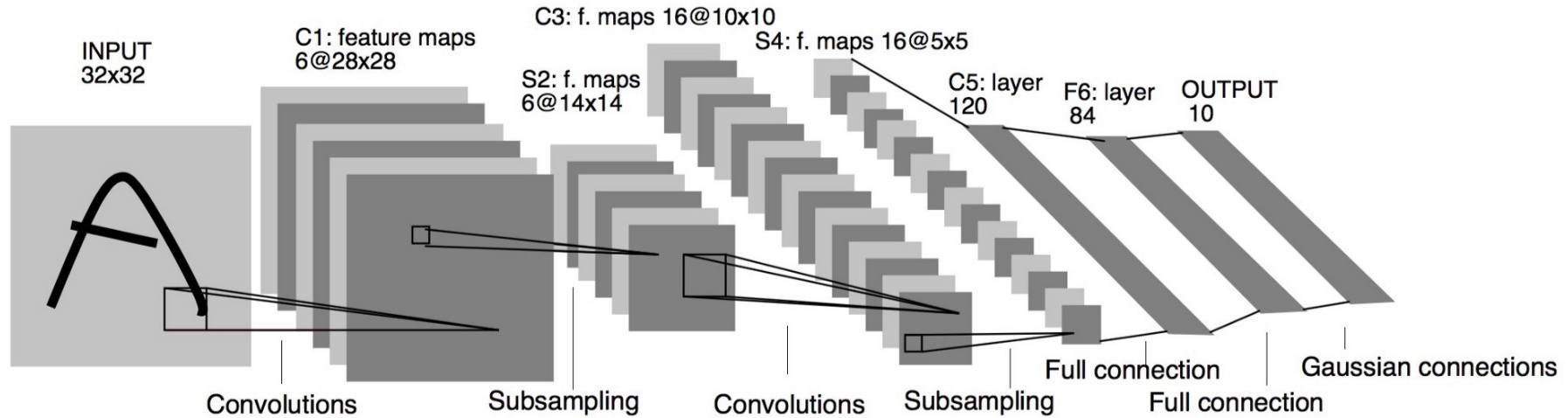
Human annotation: binary (“apple” or “not apple”)

ILSVRC Challenge



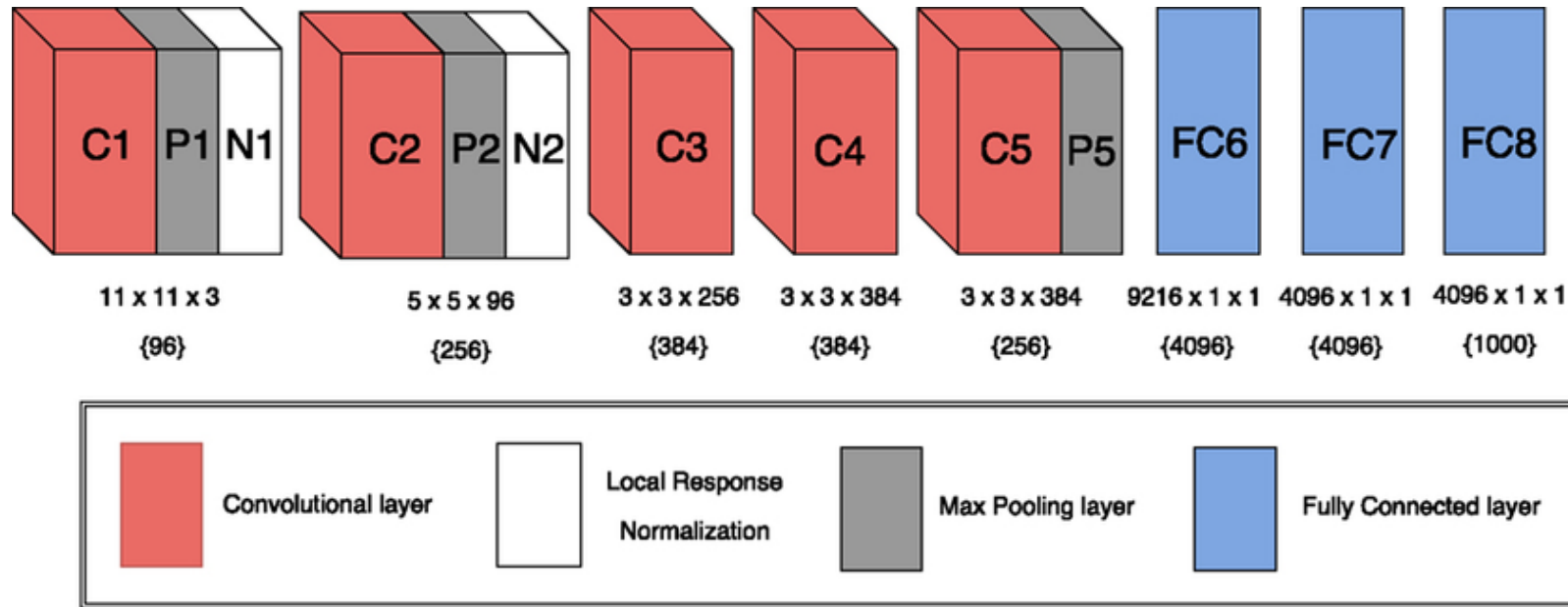
- **AlexNet (2012): First CNN for ILSVRC (16.4%)**
 - 8 layers
 - 61 million parameters
- **ZFNet (2013): 16.4% to 11.7%**
 - 8 layers
 - More filters, denser stride.
- **VGGNet (2014): 11.7% to 7.3%**
 - Beautifully uniform:
3x3 conv, stride 1, pad 1, 2x2 max pool
 - 16 layers
 - 138 million parameters
- **GoogLeNet (2014): 11.7% to 6.7%**
 - Inception module
 - 22 layers
 - 5 million parameters
(throw away FC layers)
- **ResNet (2015): 6.7% to 3.57%**
 - More layers = better performance
 - 152 layers
- **CUIImage (2016): 3.57% to 2.99%**
 - Ensemble of 6 models

LeNet5 (1994)



- The very first CNN, by Yann LeCun (1994)
- No GPU computation, non-linearity = sigmoid / tanh
- Insight:
 - Image features are distributed across the entire image
 - Convolutions with learnable features

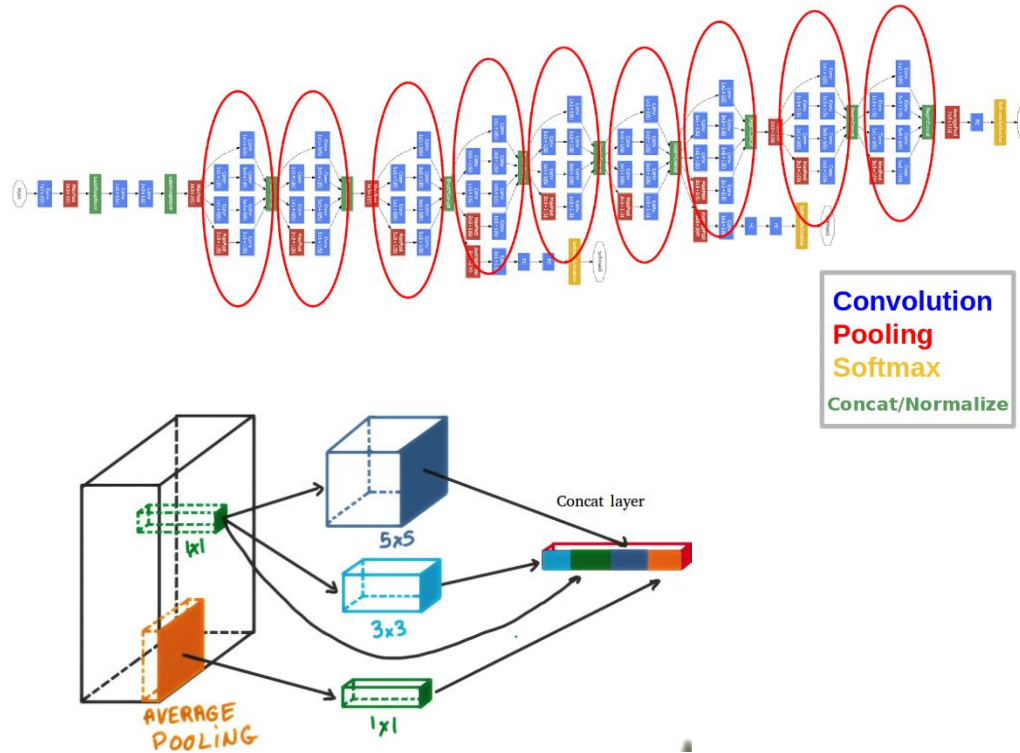
AlexNet (2012)



- Extension of LeNet5 to learn more complex objects and object hierarchy
- ReLU, dropout, overlapping max pooling, NVIDIA GTX 580

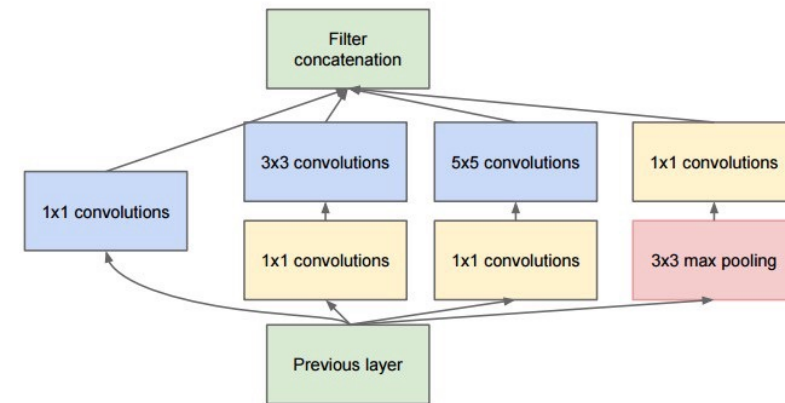
Krizhevsky et al. "Imagenet classification with deep convolutional neural networks." Advances in neural information processing systems. 2012.

GoogLeNet (2014)



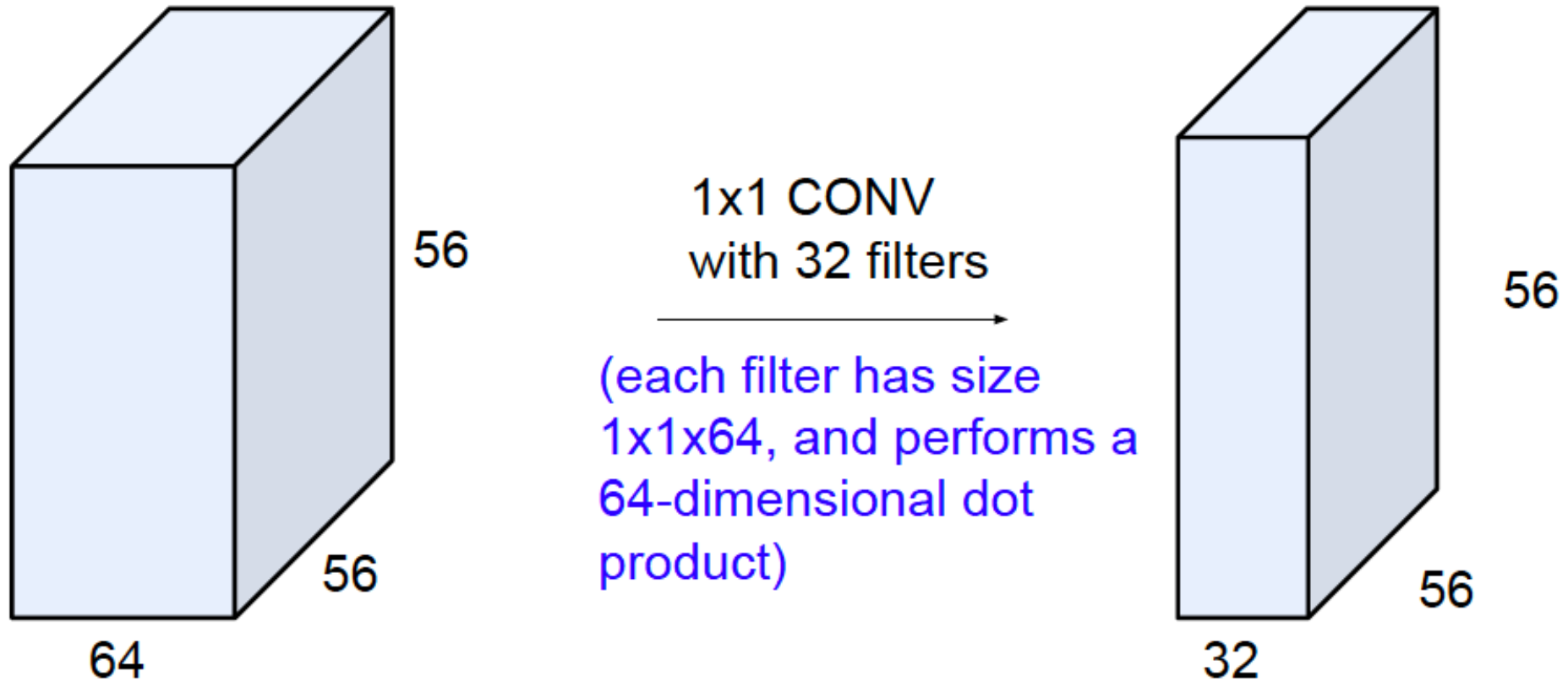
Goal: reduce no. of parameters by going deeper

“Inception” module:

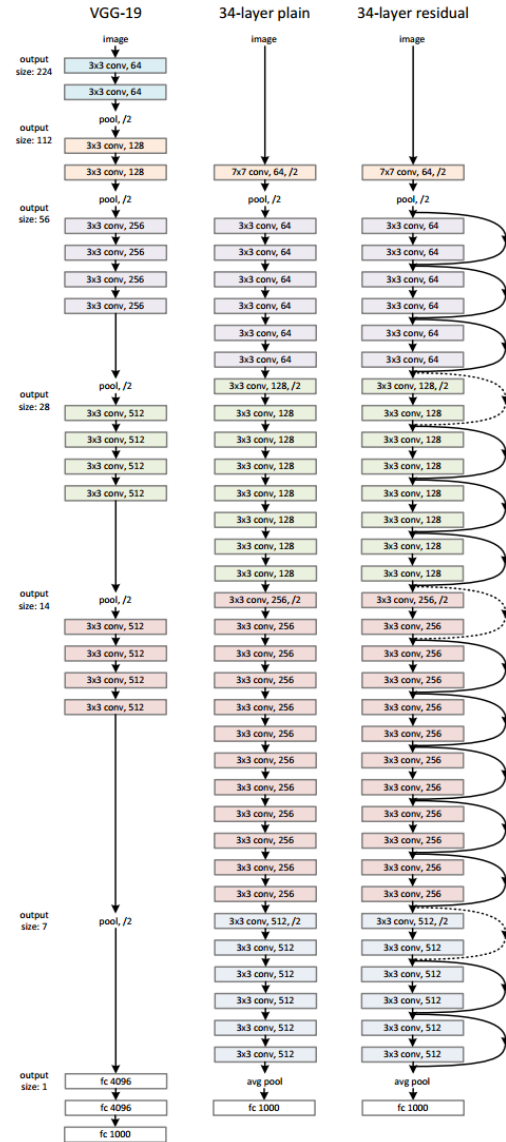


Szegedy et al. "Going deeper with convolutions." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2015.

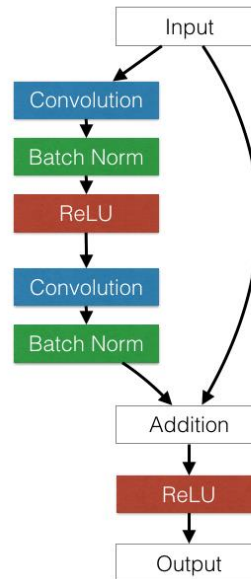
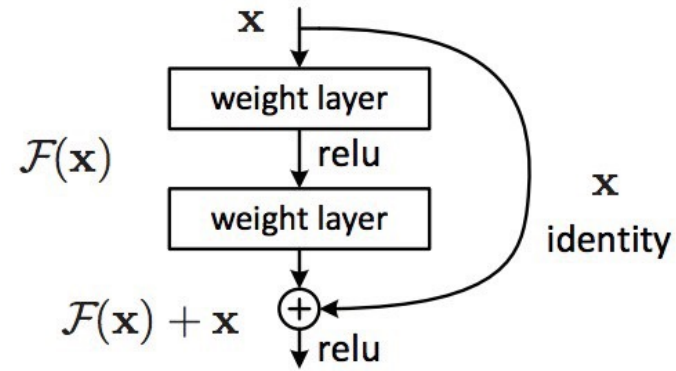
1x1 Convolution



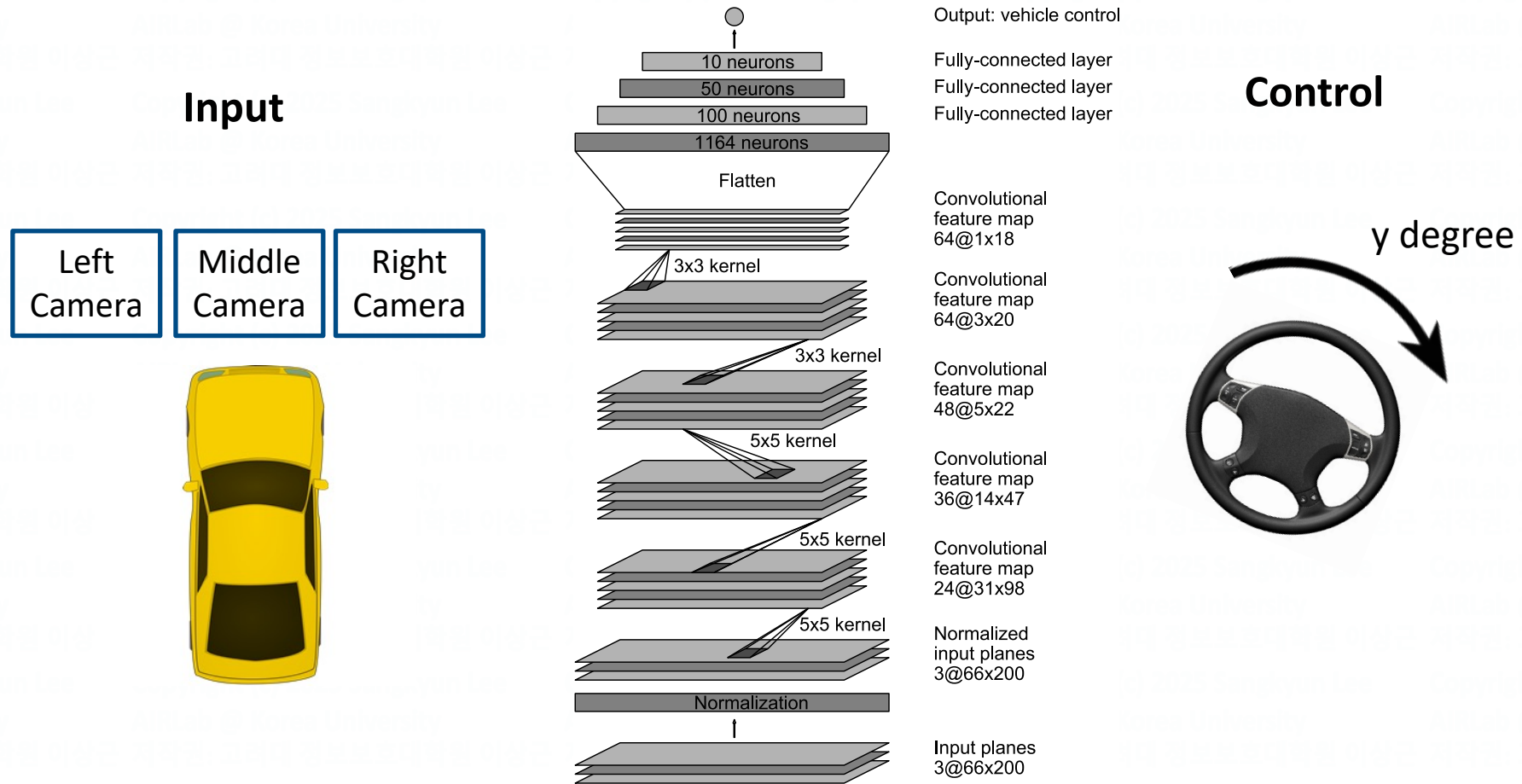
ResNet (2015)



He, Kaiming, et al. "Deep residual learning for image recognition." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016.



NVIDIA End-to-End CNN (2016)

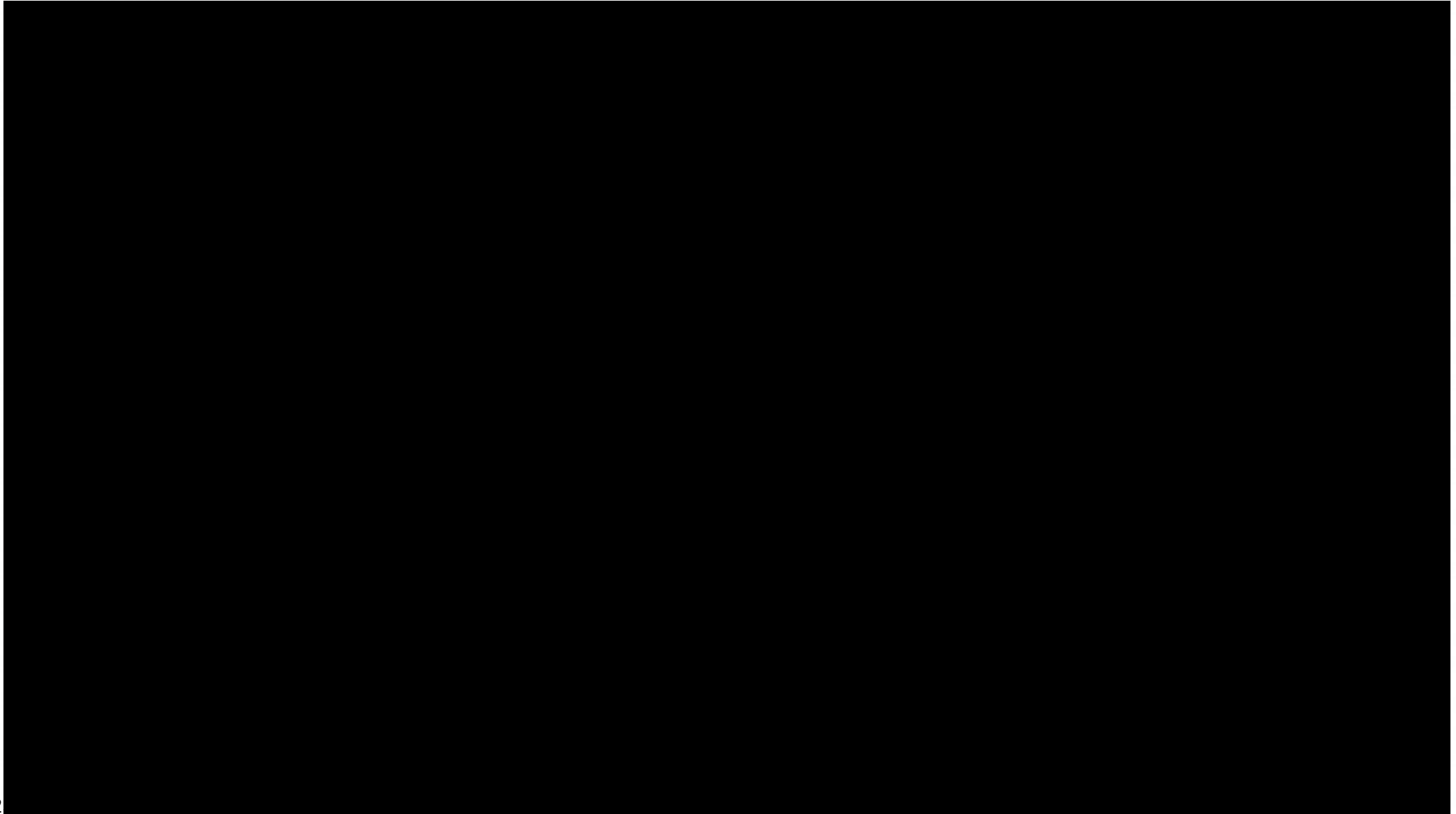


of parameters: 250,000

NVIDIA Self-Driving (2016)

Holmdel, New Jersey
February 2016

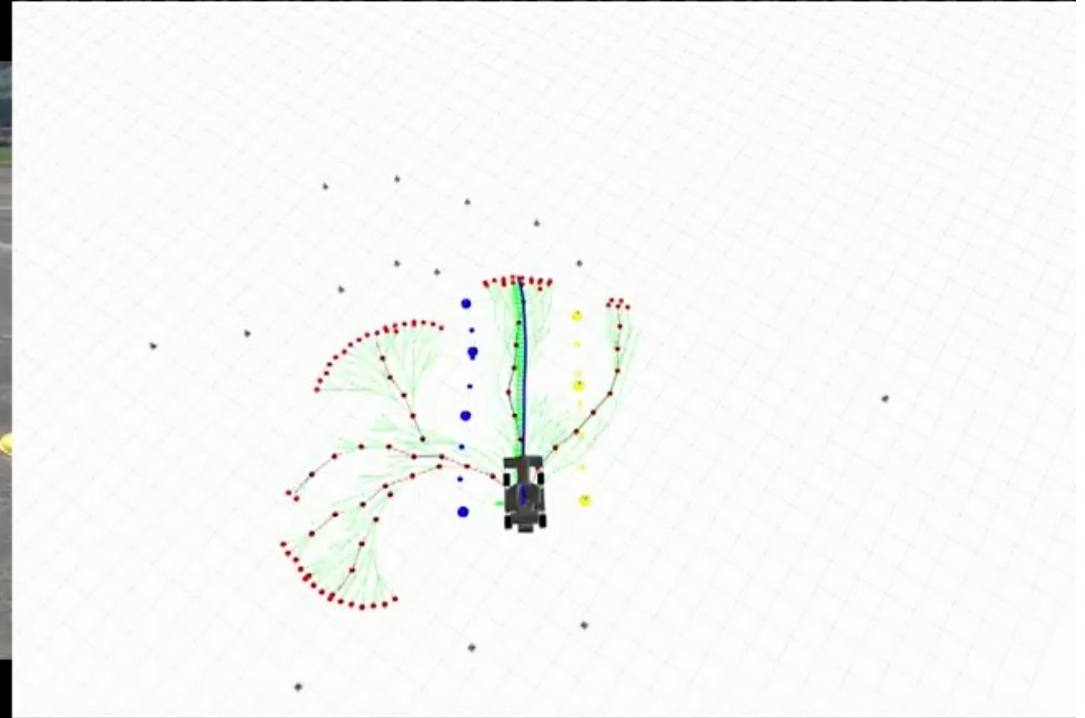
자율주행 시스템 (NVIDIA CES 2017)



DEVBOT by Roborace



Formula Students 2017 (Füela)



Thank You